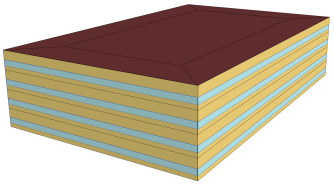
SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Commercial-Institutional
Building type	Office
Building subtype	Mid-rise (4-9 storeys)
Vintage	1983-2010

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation.

BUILDING CHARACTERISTICS

Total floor area [m²]	9,357
Conditioned floor area [m²]	9,357
Above-ground storeys	4
Climate zone	7
Weather file	Montreal-Trudeau Intl AP QC CAN CWEC2016 WMO#=716270

WINDOW-TO-WALL RATIO [%]

Average	30.0
North	30.0
East	30.0
South	30.0
West	30.0

HVAC SYSTEM

Heating	Electric boiler (hot water) + Hydronic baseboards / radiators
Cooling	Water-cooled chiller
Air distribution	VAV with reheat (1 air loop)
Heat recovery	None

ENVELOPE & INTERNAL LOADS

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²K]	0.44
Roof [W/m²K]	0.21
Exposed floors / slabs [W/m²K]	2.94
Infiltration [m³/h/m² ext. surface]	0.82

GLAZING

Glazing U-factor [W/m²K]	2.96
SHGC	0.40

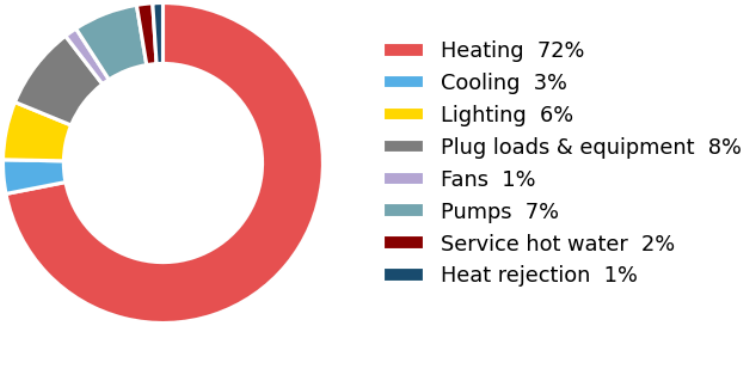
INTERNAL LOADS (MODEL INPUTS)

Occupant density [m²/person]	14.0
Lighting power density [W/m²]	6.7
Equipment power density [W/m²]	7.5

ANNUAL END USES [kWh/year]

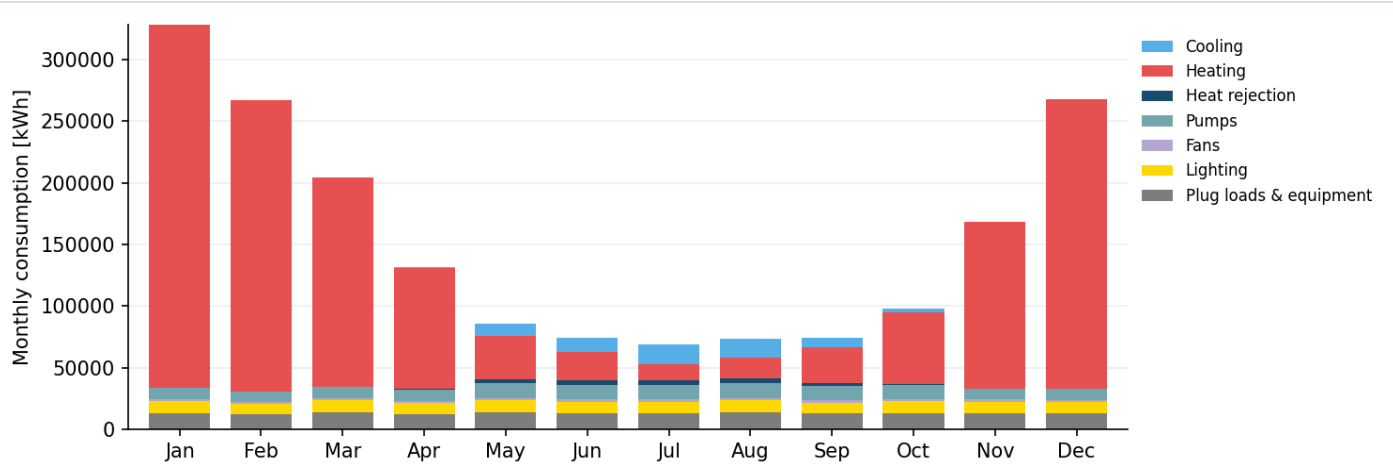
END USE	ELECTRICITY	DISTRICT HEATING	TOTAL	kWh/m²
Heating	1,346,320	0	1,346,320	143.9
Cooling	63,978	0	63,978	6.8
Interior lighting	110,083	0	110,083	11.8
Plug loads & equipment	156,783	0	156,783	16.8
Fans	23,878	0	23,878	2.6
Pumps	121,972	0	121,972	13.0
Heat rejection	19,306	0	19,306	2.1
Service hot water	0	30,025	30,025	3.2
Total			**1,872,342**	**200.1**

SHARE OF ANNUAL CONSUMPTION



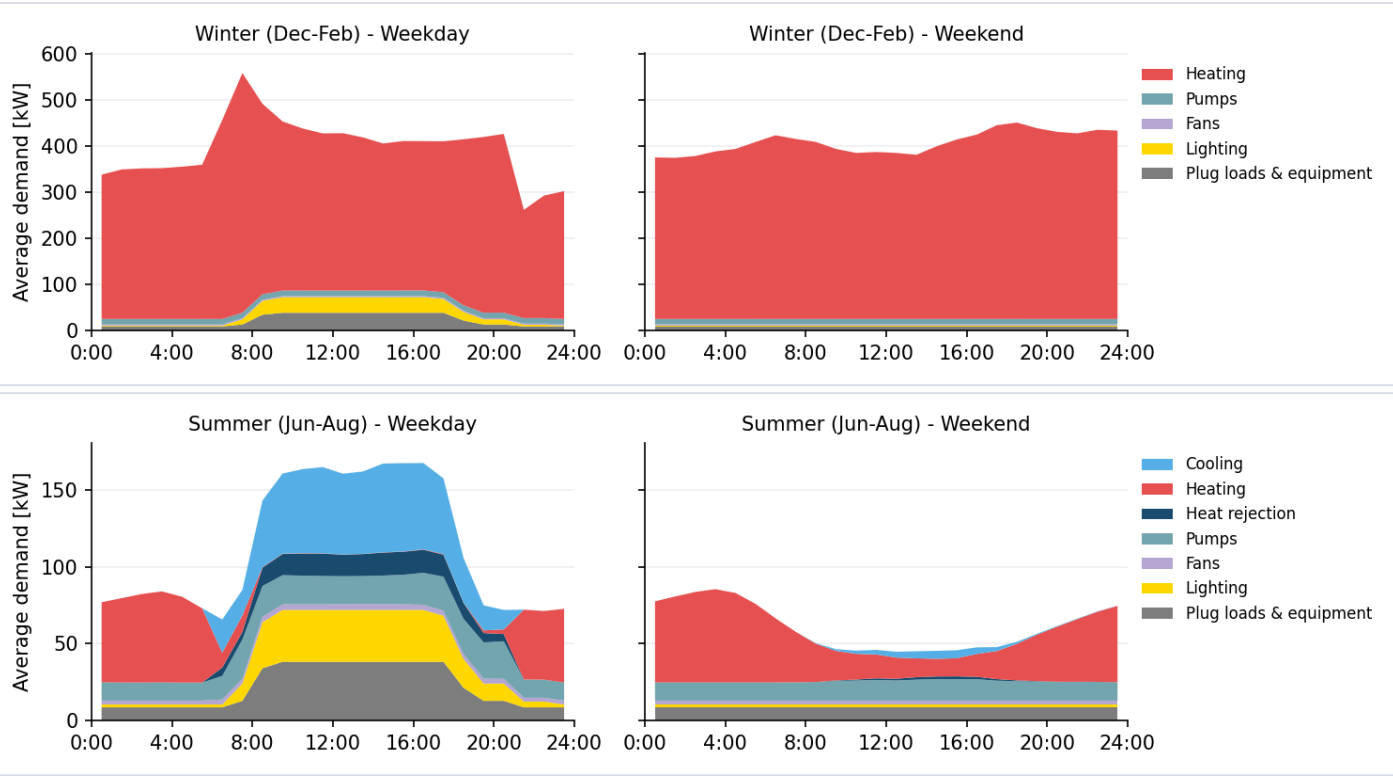
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS	OTHER FUELS	PEAK ELECTRIC DEMAND
200 kWh/m²/yr (total area)	1,872,342 kWh/yr	1,842,320 kWh/yr	0 kWh/yr	30,025 kWh/yr	1,351 kW · 144.4 W/m²

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption by end use, all fuels combined.

TYPICAL DAY PROFILES — WEEKDAY VS. WEEKEND



Average hourly demand by end use for winter and summer. Weekdays (Monday to Friday) are shown on the left and weekends and holidays on the right, showing the office occupancy schedule and HVAC set-back.

NOTES & ASSUMPTIONS

- Archetype energy model simulated with OpenStudio/EnergyPlus. Values are model outputs for the reference weather year and are not measured data.
- Energy use intensity is normalized by the total floor area (9,357 m²). Offices are not normalized per unit.
- Archetype: Mid-rise office (4-9 storeys), built 1983-2010.
- HVAC entries marked "(typical)" are not modelled explicitly and come from the archetype assumption table.
- Peak electric demand occurs on 24-DEC-06:20.